KBTU KAZAKH-BRITISH TECHNICAL UNIVERSITY

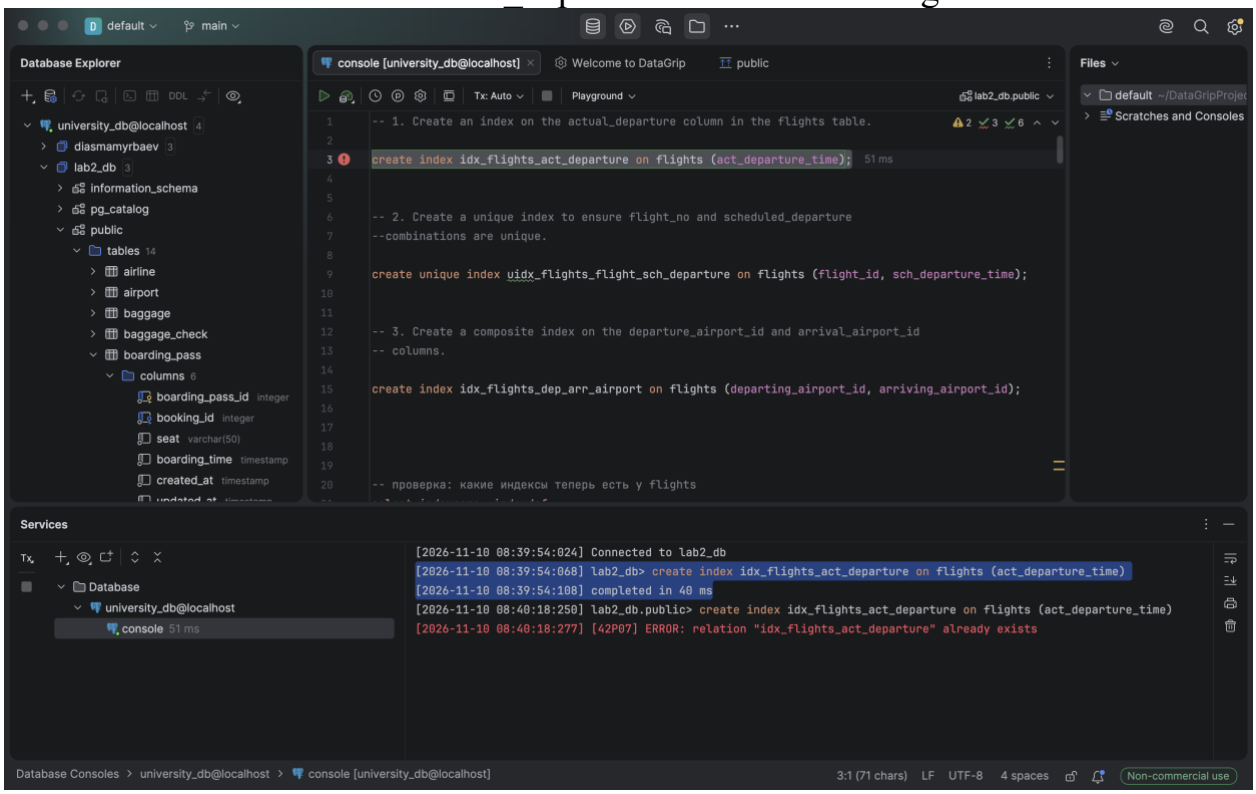
Course name: Databases

Laboratory work 7

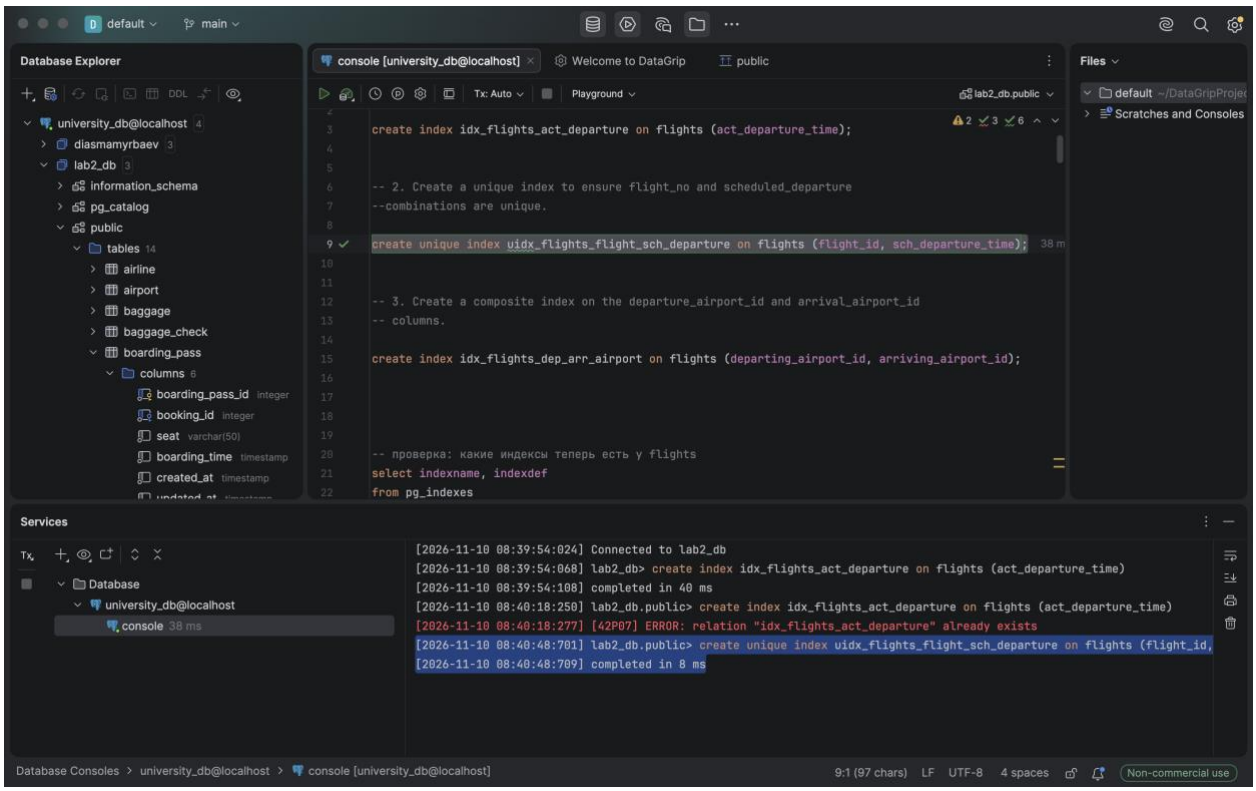
Submission format: please work with the database from the prev lab work and provide full-page screenshots that covers the code and results of each task in the PDF format, also provide sql files.

Required tasks:

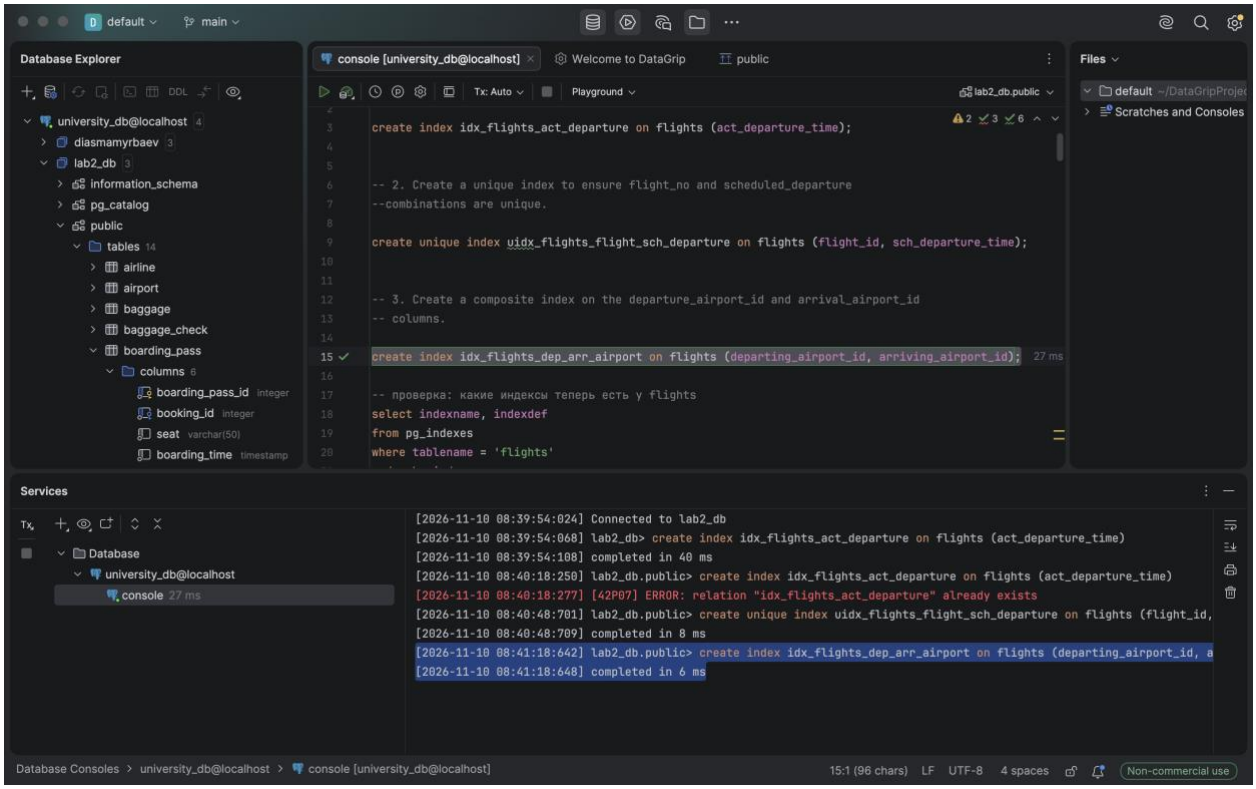
1. Create an index on the actual_departure column in the flights table.

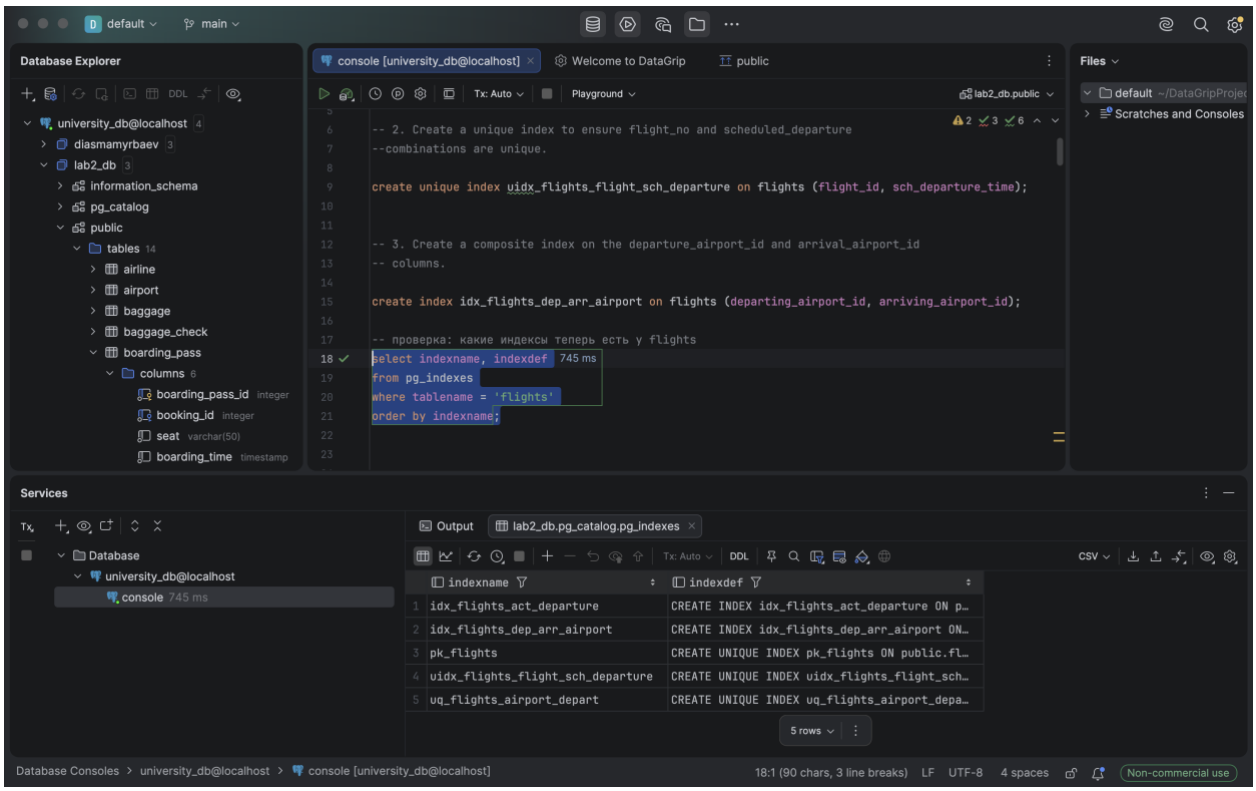


2. Create a unique index to ensure flight_no and scheduled_departure combinations are unique.

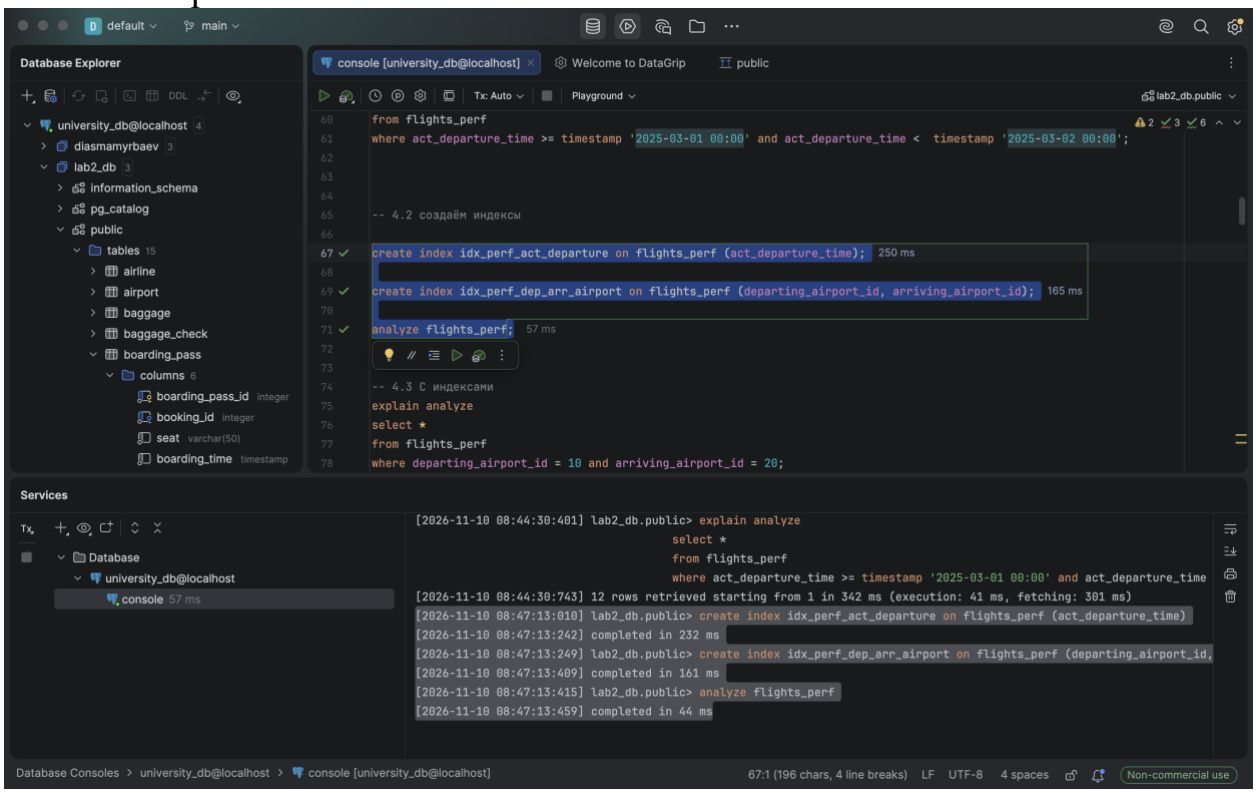


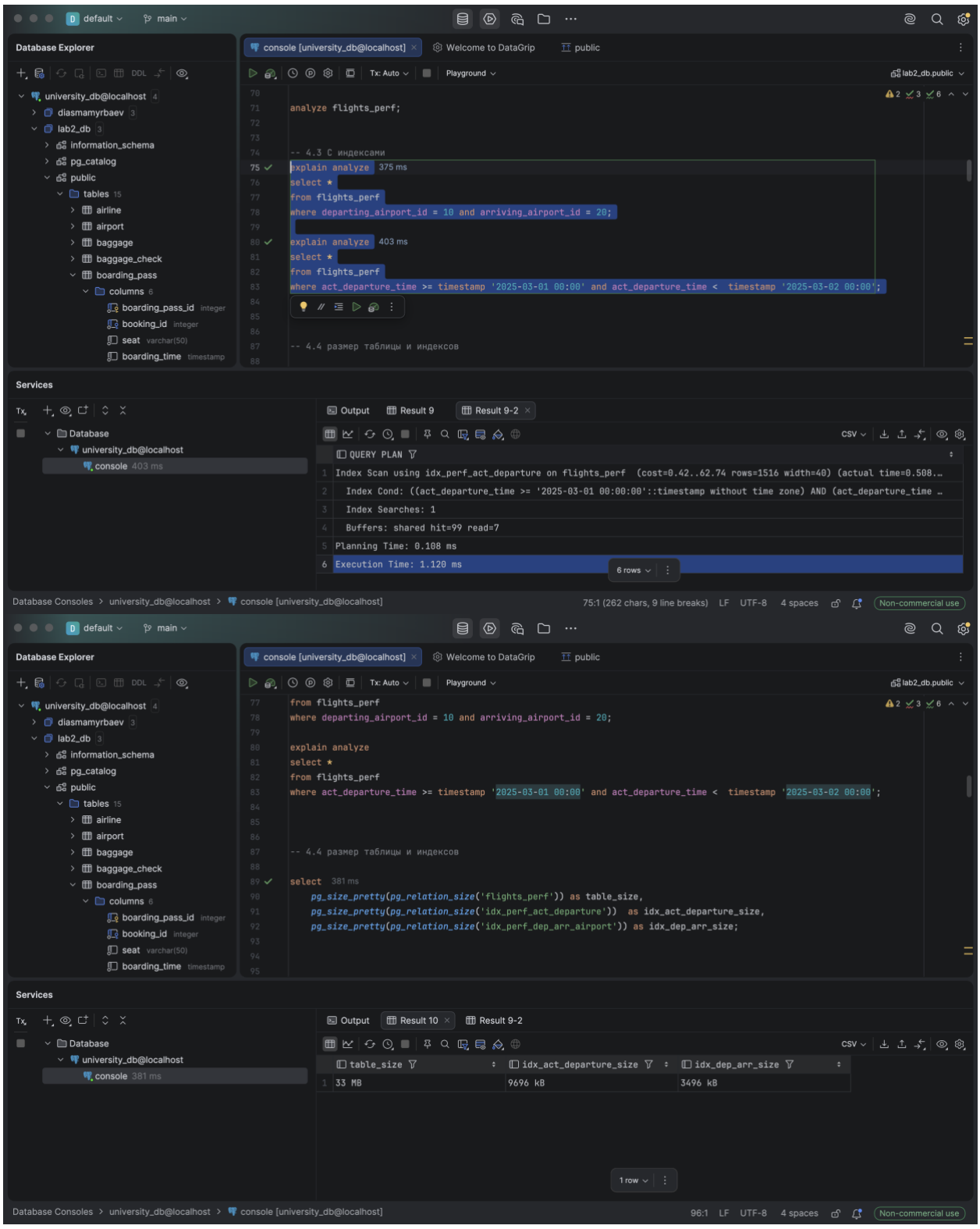
3. Create a composite index on the departure_airport_id and arrival_airport_id columns.



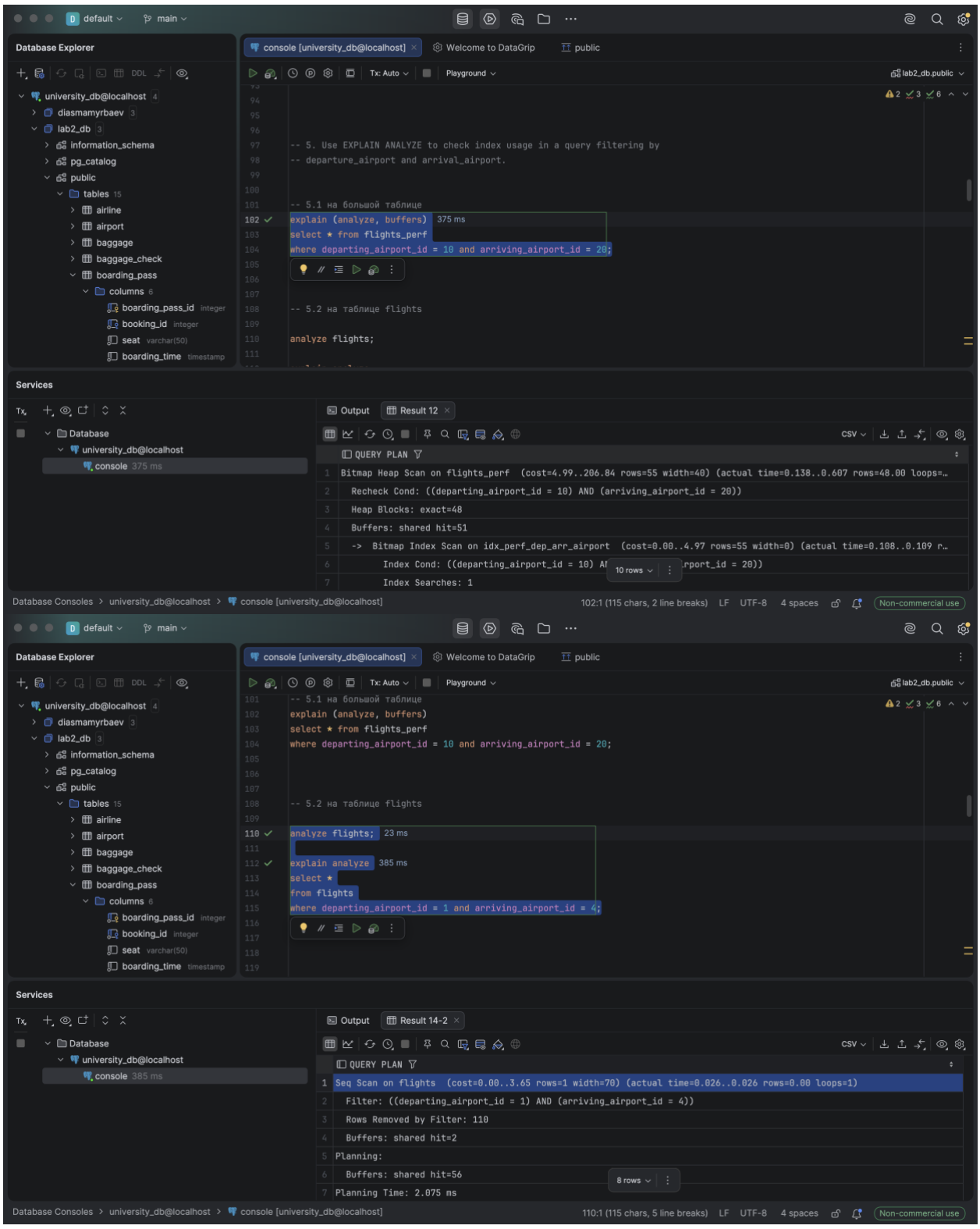


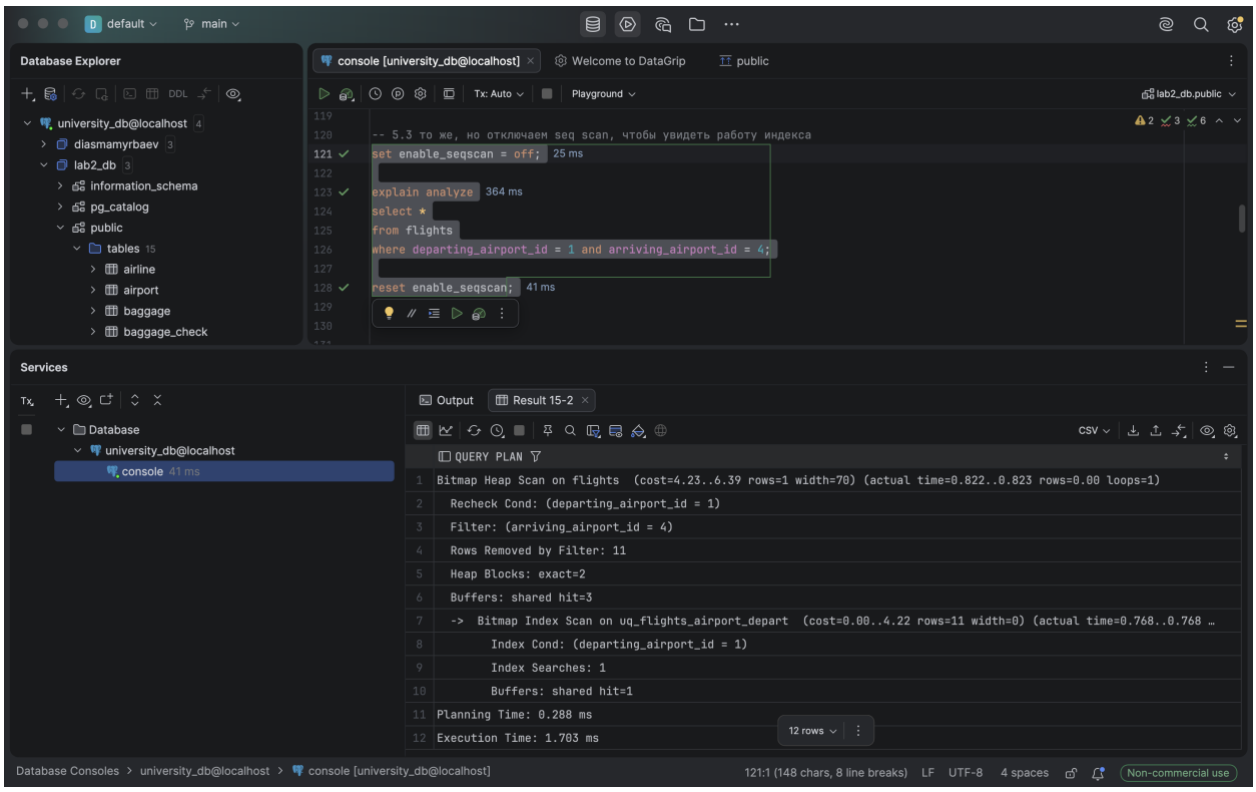
4. Evaluate the difference in query performance with and without indexes. Measure performance differences.



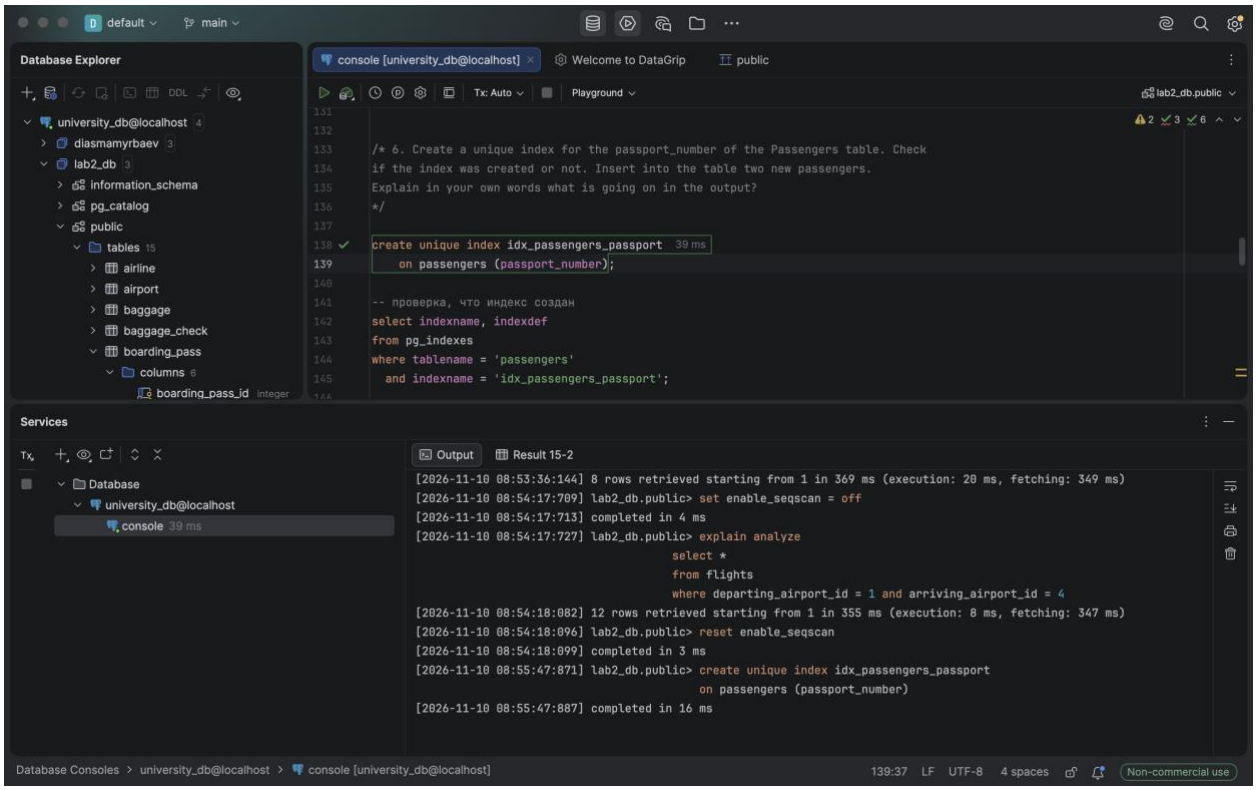


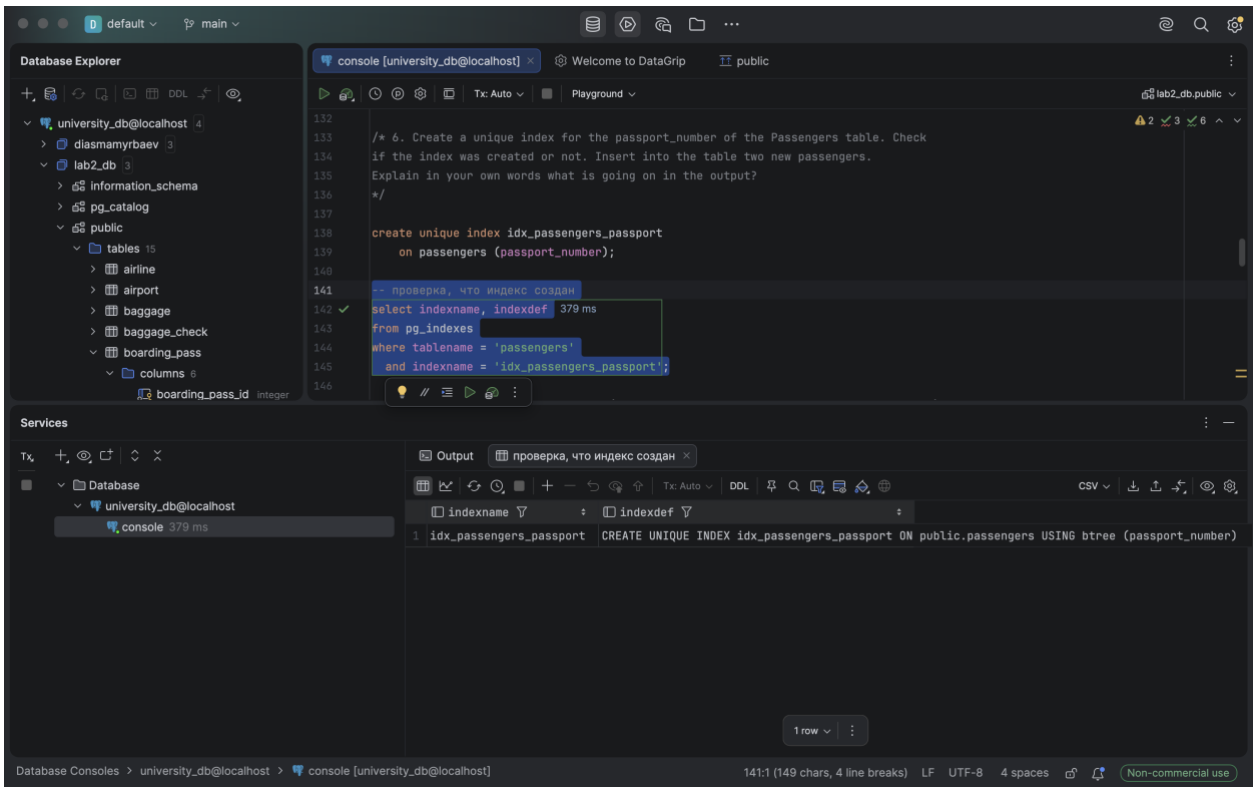
5. Use EXPLAIN ANALYZE to check index usage in a query filtering by departure_airport and arrival_airport.

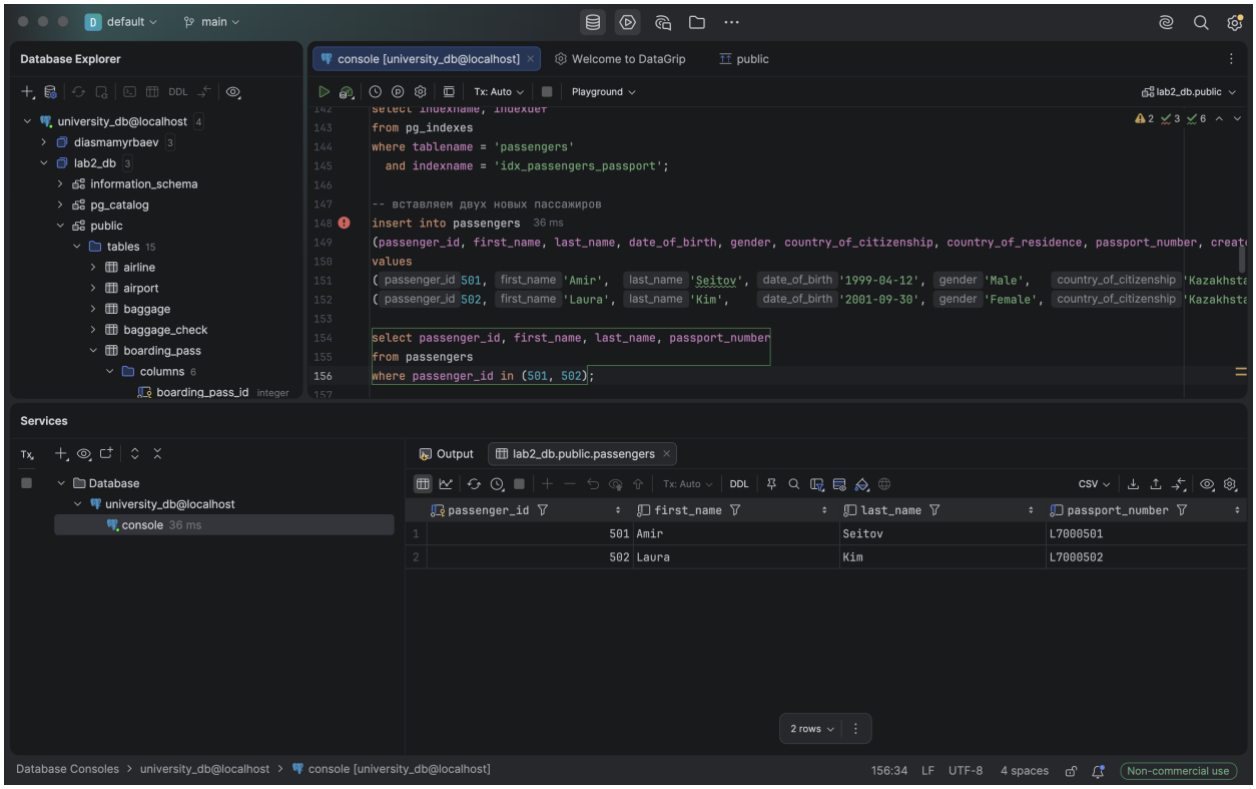

default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
boarding_pass_id integer
booking_id integer
seat varchar(50)
boarding_time timestamp
-- 5. Use EXPLAIN ANALYZE to check index usage in a query filtering by
-- departure_airport and arrival_airport.
-- 5.1 на большой таблице
explain (analyze, buffers) 375 ms
select * from flights_perf
where departing_airport_id = 10 and arriving_airport_id = 20;
-- 5.2 на таблице flights
analyze flights;
Services
Database
university_db@localhost
console 375 ms
Output
Result 12
CSV
QUERY PLAN
Bitmap Heap Scan on flights_perf (cost=4.99..206.84 rows=55 width=40) (actual time=0.138..0.607 rows=48.00 loops=…
Recheck Cond: ((departing_airport_id = 10) AND (arriving_airport_id = 20))
Heap Blocks: exact=48
Buffers: shared hit=51
-> Bitmap Index Scan on idx_perf_dep_arr_airport (cost=0.00..4.97 rows=55 width=0) (actual time=0.108..0.109 r…
Index Cond: ((departing_airport_id = 10) AI
rport_id = 20))
Index Searches: 1
10 rows
Database Consoles
university_db@localhost
console [university_db@localhost]
102:1 (115 chars, 2 line breaks)
LF
UTF-8
4 spaces
Non-commercial use
default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
boarding_pass_id integer
booking_id integer
seat varchar(50)
boarding_time timestamp
-- 5.1 на большой таблице
explain (analyze, buffers)
select * from flights_perf
where departing_airport_id = 10 and arriving_airport_id = 20;
-- 5.2 на таблице flights
analyze flights; 23 ms
explain analyze 385 ms
select *
from flights
where departing_airport_id = 1 and arriving_airport_id = 4;
Services
Database
university_db@localhost
console 385 ms
Output
Result 14-2
CSV
QUERY PLAN
Seq Scan on flights (cost=0.00..3.65 rows=1 width=70) (actual time=0.026..0.026 rows=0.00 loops=1)
Filter: ((departing_airport_id = 1) AND (arriving_airport_id = 4))
Rows Removed by Filter: 110
Buffers: shared hit=2
Planning:
Buffers: shared hit=56
Planning Time: 2.075 ms
8 rows
Database Consoles
university_db@localhost
console [university_db@localhost]
110:1 (115 chars, 5 line breaks)
LF
UTF-8
4 spaces
Non-commercial use

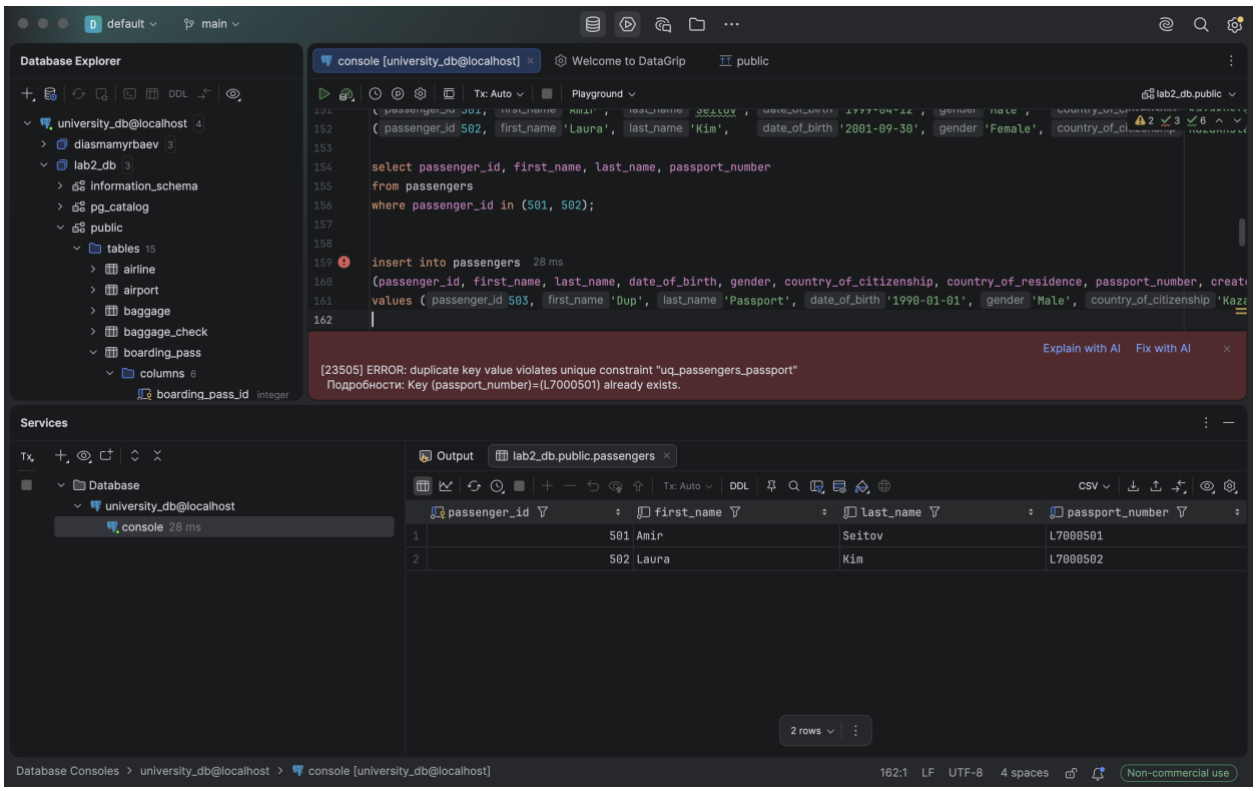


6. Create a unique index for the passport_number of the Passengers table. Check if the index was created or not. Insert into the table two new passengers. Explain in your own words what is going on in the output?

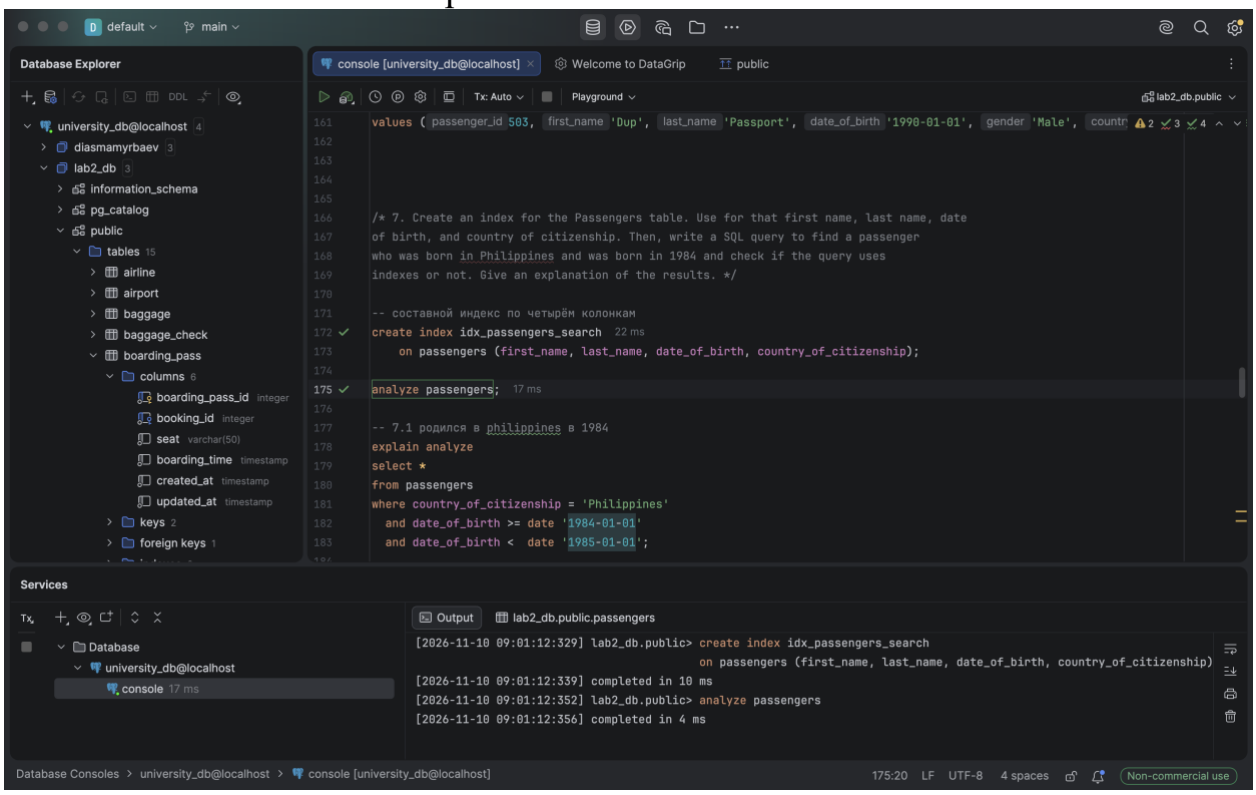


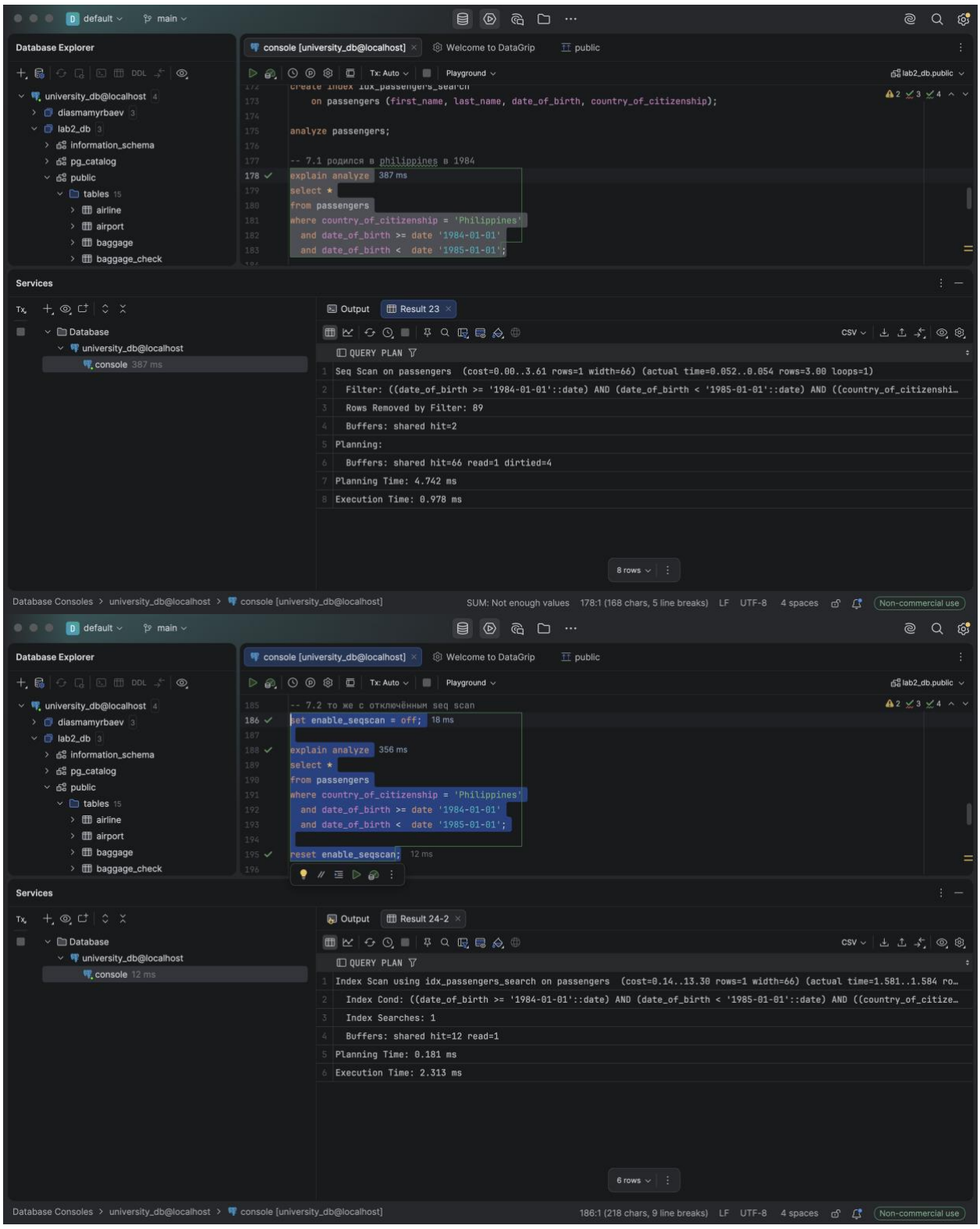

Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
boarding_pass_id integer
/* 6. Create a unique index for the passport_number of the Passengers table. Check
if the index was created or not. Insert into the table two new passengers.
Explain in your own words what is going on in the output?
*/
create unique index idx_passengers_passport
on passengers (passport_number);
-- проверка, что индекс создан
select indexname, indexdef 379 ms
from pg_indexes
where tablename = 'passengers'
and indexname = 'idx_passengers_passport';
Services
Database
university_db@localhost
console 379 ms
Output
проверка, что индекс создан
indexname
indexdef
idx_passengers_passport
CREATE UNIQUE INDEX idx_passengers_passport ON public.passengers USING btree (passport_number)
1 row
Database Consoles
university_db@localhost
console [university_db@localhost]
141:1 (149 chars, 4 line breaks)
LF
UTF-8
4 spaces
Non-commercial use

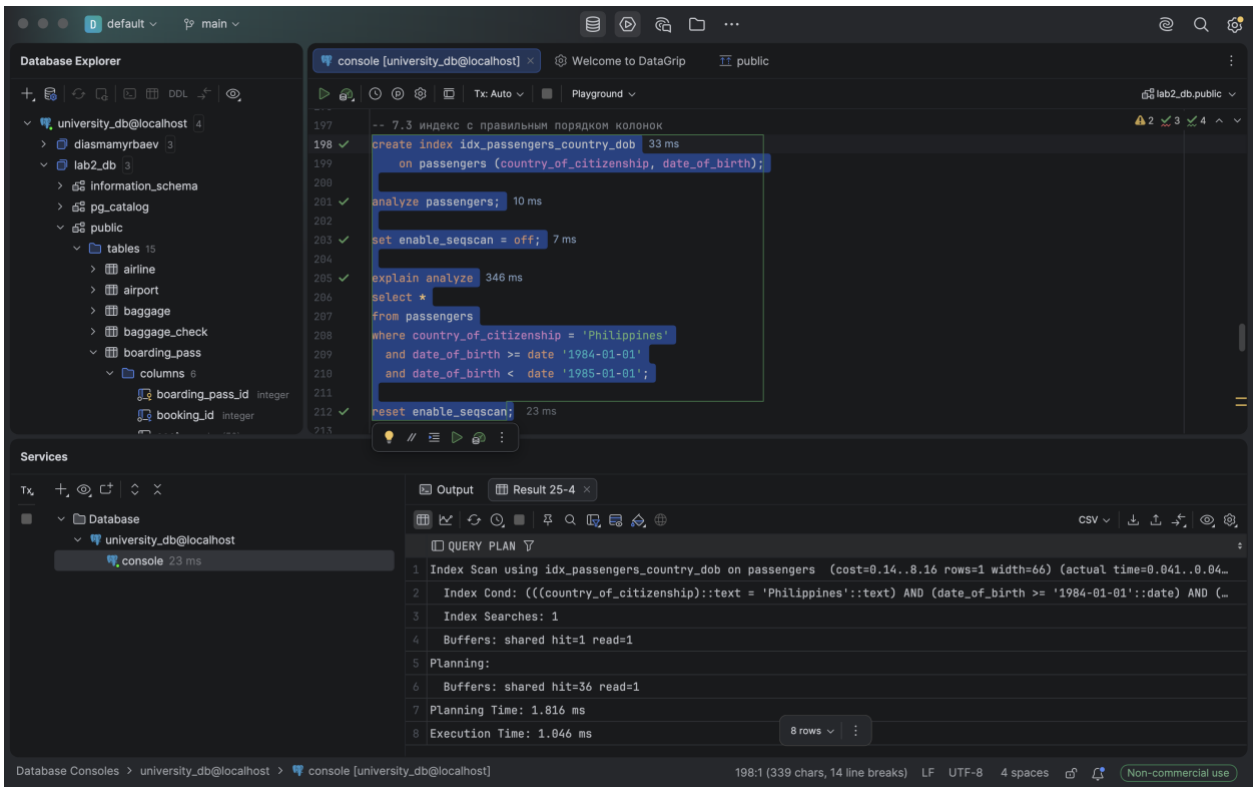

Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
boarding_pass_id integer
from pg_indexes
where tablename = 'passengers'
and indexname = 'idx_passengers_passport';
-- вставляем двух новых пассажиров
insert into passengers 36 ms
(passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, creat
values
(passenger_id 501, first_name 'Amir', last_name 'Seitov', date_of_birth '1999-04-12', gender 'Male', country_of_citizenship 'Kazakhst
(passenger_id 502, first_name 'Laura', last_name 'Kim', date_of_birth '2001-09-30', gender 'Female', country_of_citizenship 'Kazakhst
select passenger_id, first_name, last_name, passport_number
from passengers
where passenger_id in (501, 502);
Services
Database
university_db@localhost
console 36 ms
Output
lab2_db.public.passengers
passenger_id
first_name
last_name
passport_number
501 Amir Seitov L7000501
502 Laura Kim L7000502
2 rows
Database Consoles
university_db@localhost
console [university_db@localhost]
156:34
LF
UTF-8
4 spaces
Non-commercial use



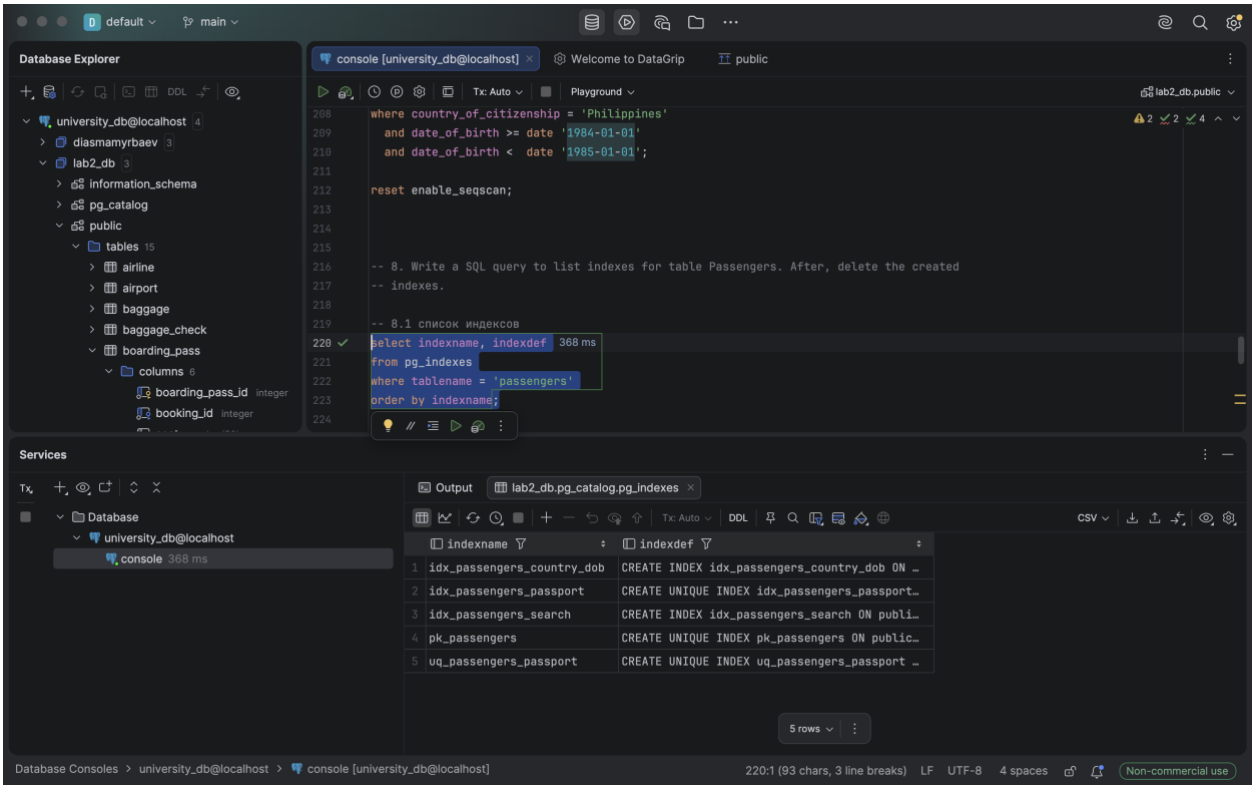
7. Create an index for the Passengers table. Use for that first name, last name, date of birth, and country of citizenship. Then, write a SQL query to find a passenger who was born in Philippines and was born in 1984 and check if the query uses indexes or not. Give an explanation of the results.

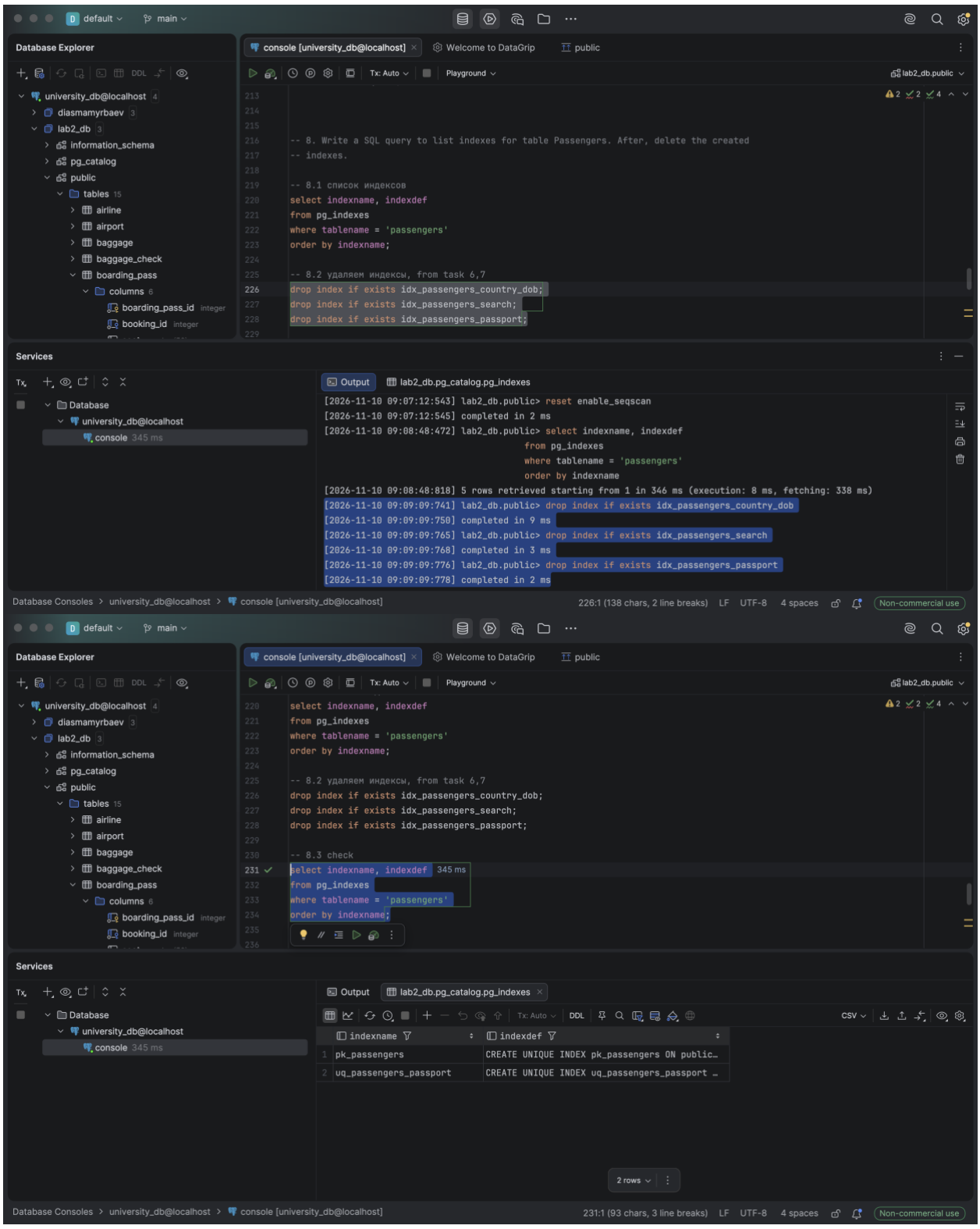



default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
create index idx_passengers_search
on passengers (first_name, last_name, date_of_birth, country_of_citizenship);
analyze passengers;
-- 7.1 родился в philippines в 1984
explain analyze 387 ms
select *
from passengers
where country_of_citizenship = 'Philippines'
and date_of_birth >= date '1984-01-01'
and date_of_birth < date '1985-01-01';
Services
Database
university_db@localhost
console 387 ms
Output
Result 23
CSV
QUERY PLAN
Seq Scan on passengers (cost=0.00..3.61 rows=1 width=66) (actual time=0.052..0.054 rows=3.00 loops=1)
Filter: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citizenshi…
Rows Removed by Filter: 89
Buffers: shared hit=2
Planning:
Buffers: shared hit=66 read=1 dirtied=4
Planning Time: 4.742 ms
Execution Time: 0.978 ms
8 rows
Database Consoles
university_db@localhost
console [university_db@localhost]
SUM: Not enough values
178:1 (168 chars, 5 line breaks)
LF
UTF-8
4 spaces
Non-commercial use
default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
-- 7.2 то же с отключённым seq scan
set enable_seqscan = off; 18 ms
explain analyze 356 ms
select *
from passengers
where country_of_citizenship = 'Philippines'
and date_of_birth >= date '1984-01-01'
and date_of_birth < date '1985-01-01';
reset enable_seqscan; 12 ms
Services
Database
university_db@localhost
console 12 ms
Output
Result 24-2
CSV
QUERY PLAN
Index Scan using idx_passengers_search on passengers (cost=0.14..13.30 rows=1 width=66) (actual time=1.581..1.584 ro…
Index Cond: ((date_of_birth >= '1984-01-01'::date) AND (date_of_birth < '1985-01-01'::date) AND ((country_of_citize…
Index Searches: 1
Buffers: shared hit=12 read=1
Planning Time: 0.181 ms
Execution Time: 2.313 ms
6 rows
Database Consoles
university_db@localhost
console [university_db@localhost]
186:1 (218 chars, 9 line breaks)
LF
UTF-8
4 spaces
Non-commercial use

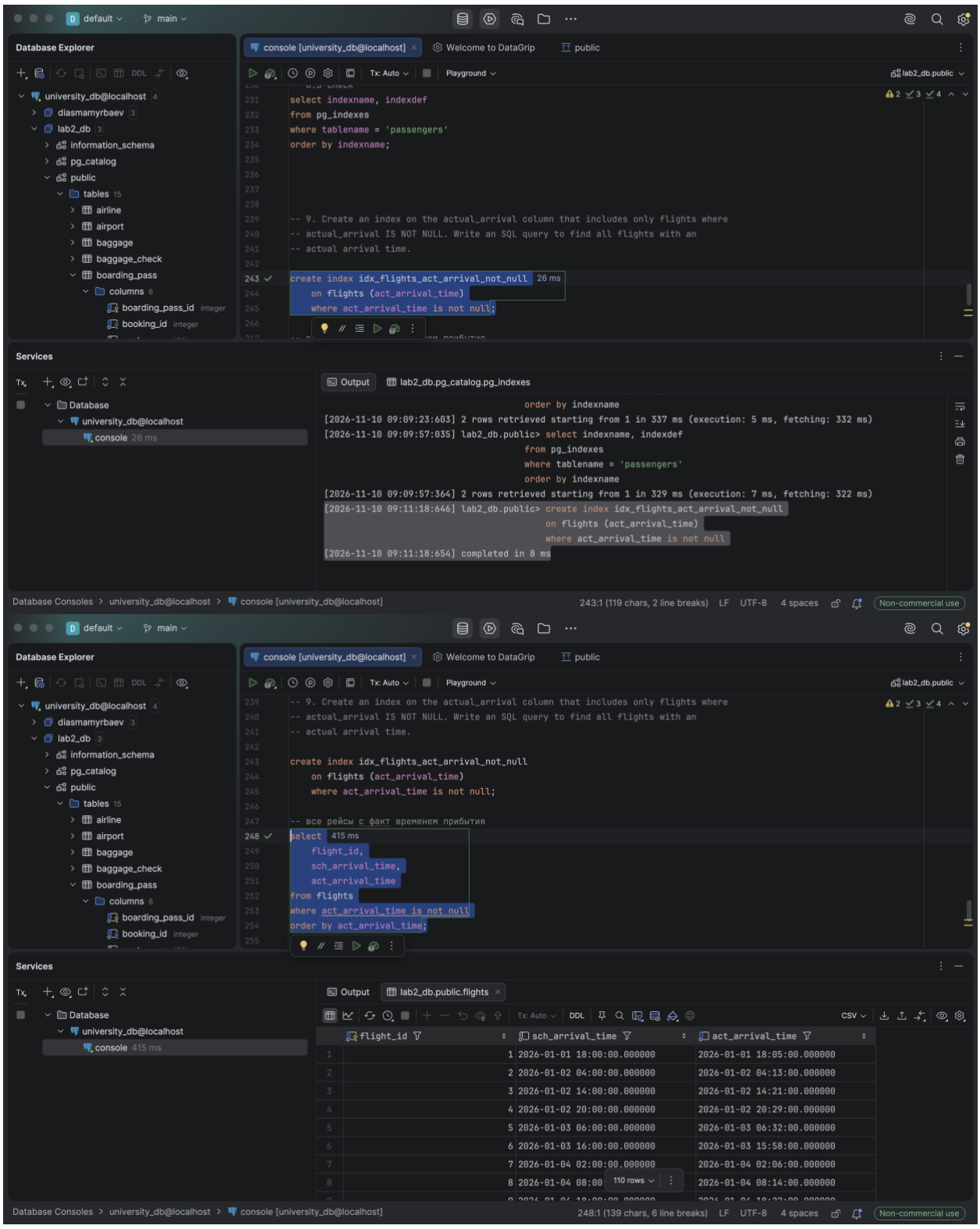


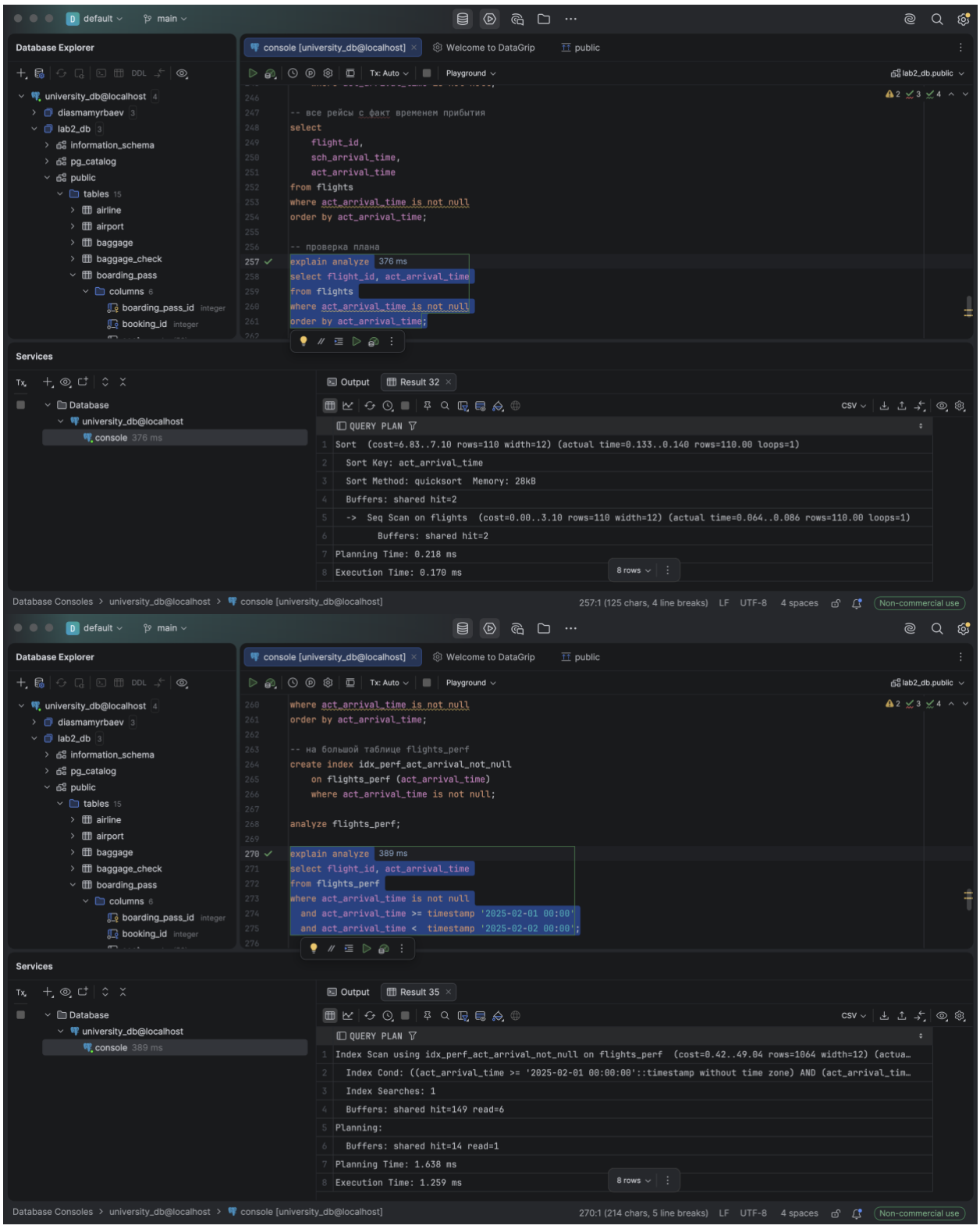
8. Write a SQL query to list indexes for table Passengers. After, delete the created indexes.



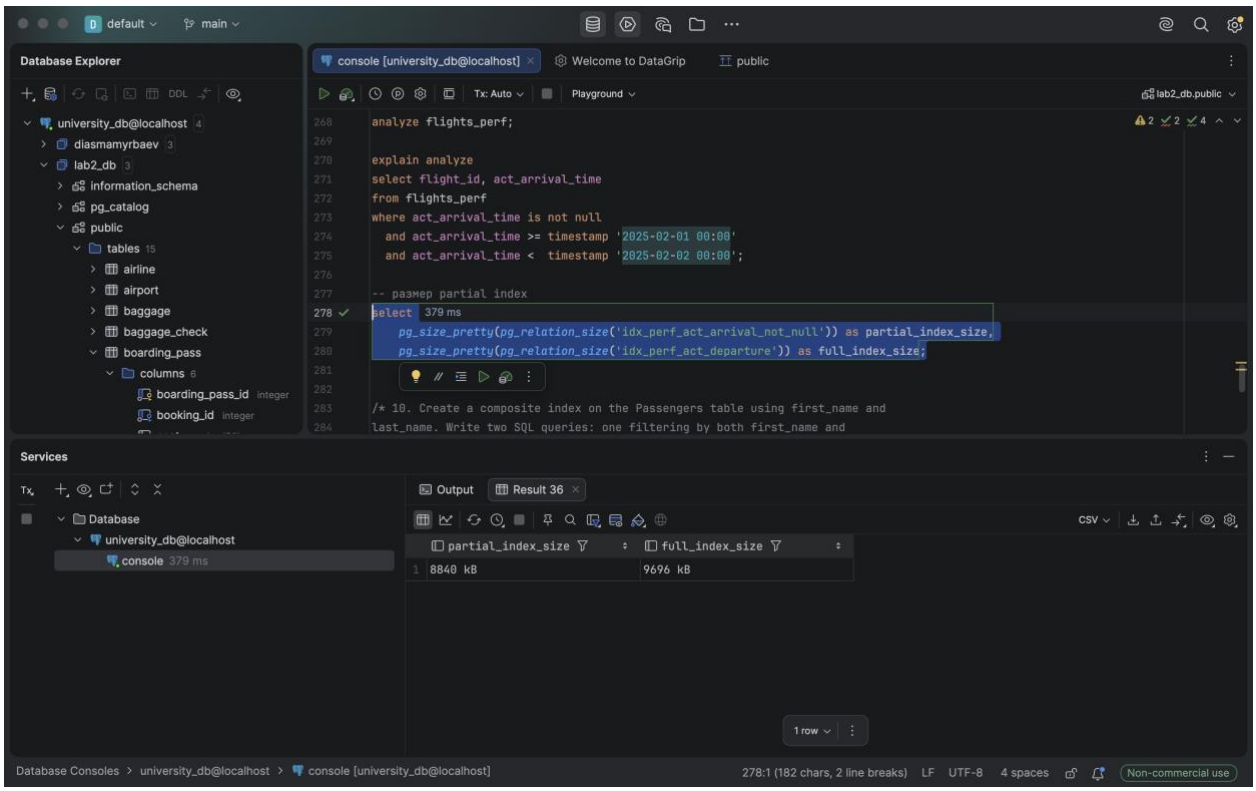


9. Create an index on the actual_arrival column that includes only flights where actual_arrival IS NOT NULL. Write an SQL query to find all flights with an actual arrival time.

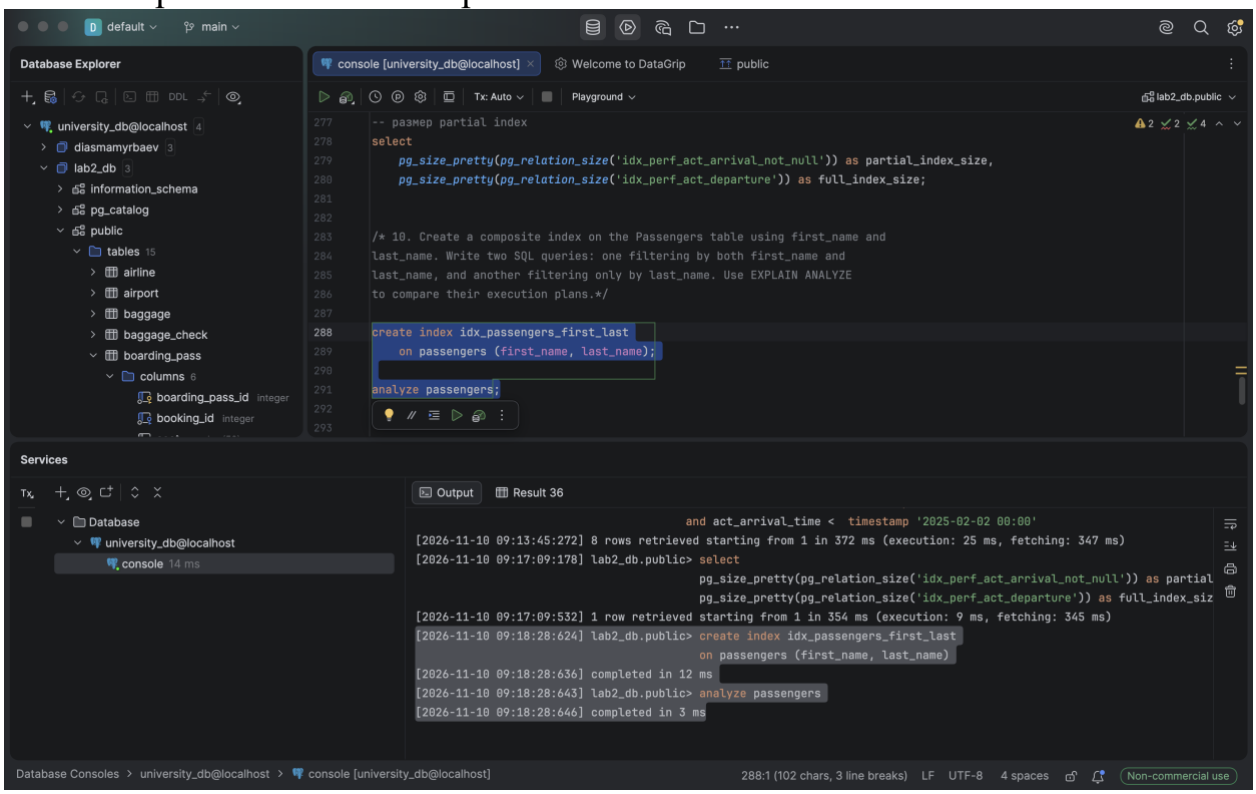

default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
boarding_pass_id integer
booking_id integer
select indexname, indexdef
from pg_indexes
where tablename = 'passengers'
order by indexname;
-- 9. Create an index on the actual_arrival column that includes only flights where
-- actual_arrival IS NOT NULL. Write an SQL query to find all flights with an
-- actual arrival time.
create index idx_flights_act_arrival_not_null 26 ms
on flights (act_arrival_time)
where act_arrival_time is not null;
Services
Database
university_db@localhost
console 26 ms
Output
lab2_db.pg_catalog.pg_indexes
order by indexname
[2026-11-10 09:09:23:603] 2 rows retrieved starting from 1 in 337 ms (execution: 5 ms, fetching: 332 ms)
[2026-11-10 09:09:57:035] lab2_db.public> select indexname, indexdef
from pg_indexes
where tablename = 'passengers'
order by indexname
[2026-11-10 09:09:57:364] 2 rows retrieved starting from 1 in 329 ms (execution: 7 ms, fetching: 322 ms)
[2026-11-10 09:11:18:646] lab2_db.public> create index idx_flights_act_arrival_not_null
on flights (act_arrival_time)
where act_arrival_time is not null
[2026-11-10 09:11:18:654] completed in 8 ms
Database Consoles > university_db@localhost > console [university_db@localhost]
243:1 (119 chars, 2 line breaks) LF UTF-8 4 spaces
Non-commercial use
-- все рейсы с факт временем прибытия
select 415 ms
flight_id,
sch_arrival_time,
act_arrival_time
from flights
where act_arrival_time is not null
order by act_arrival_time;
console 415 ms
Output
lab2_db.public.flights
CSV
DDL
flight_id
sch_arrival_time
act_arrival_time
1 2026-01-01 18:00:00.000000 2026-01-01 18:05:00.000000
2 2026-01-02 04:00:00.000000 2026-01-02 04:13:00.000000
3 2026-01-02 14:00:00.000000 2026-01-02 14:21:00.000000
4 2026-01-02 20:00:00.000000 2026-01-02 20:29:00.000000
5 2026-01-03 06:00:00.000000 2026-01-03 06:32:00.000000
6 2026-01-03 16:00:00.000000 2026-01-03 15:58:00.000000
7 2026-01-04 02:00:00.000000 2026-01-04 02:06:00.000000
8 2026-01-04 08:00:00.000000 2026-01-04 08:14:00.000000
110 rows
248:1 (139 chars, 6 line breaks) LF UTF-8 4 spaces
Non-commercial use

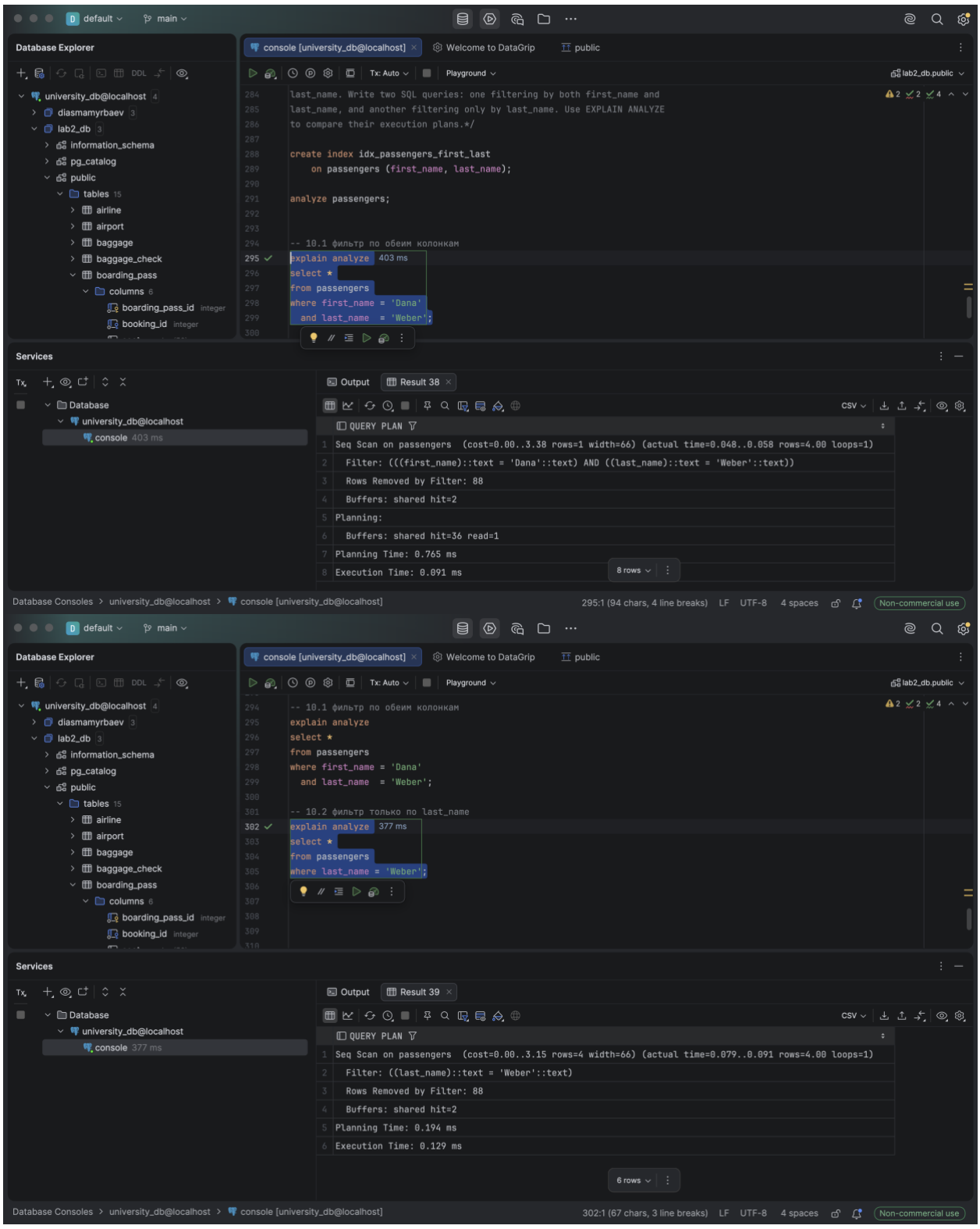

default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
boarding_pass_id integer
booking_id integer
-- все рейсы с факт временем прибытия
select
flight_id,
sch_arrival_time,
act_arrival_time
from flights
where act_arrival_time is not null
order by act_arrival_time;
-- проверка плана
explain analyze 376 ms
select flight_id, act_arrival_time
from flights
where act_arrival_time is not null
order by act_arrival_time;
Services
Database
university_db@localhost
console 376 ms
Output
Result 32
CSV
QUERY PLAN
Sort (cost=6.83..7.10 rows=110 width=12) (actual time=0.133..0.140 rows=110.00 loops=1)
Sort Key: act_arrival_time
Sort Method: quicksort Memory: 28kB
Buffers: shared hit=2
-> Seq Scan on flights (cost=0.00..3.10 rows=110 width=12) (actual time=0.064..0.086 rows=110.00 loops=1)
Buffers: shared hit=2
Planning Time: 0.218 ms
Execution Time: 0.170 ms
8 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
257:1 (125 chars, 4 line breaks)
LF
UTF-8
4 spaces
Non-commercial use

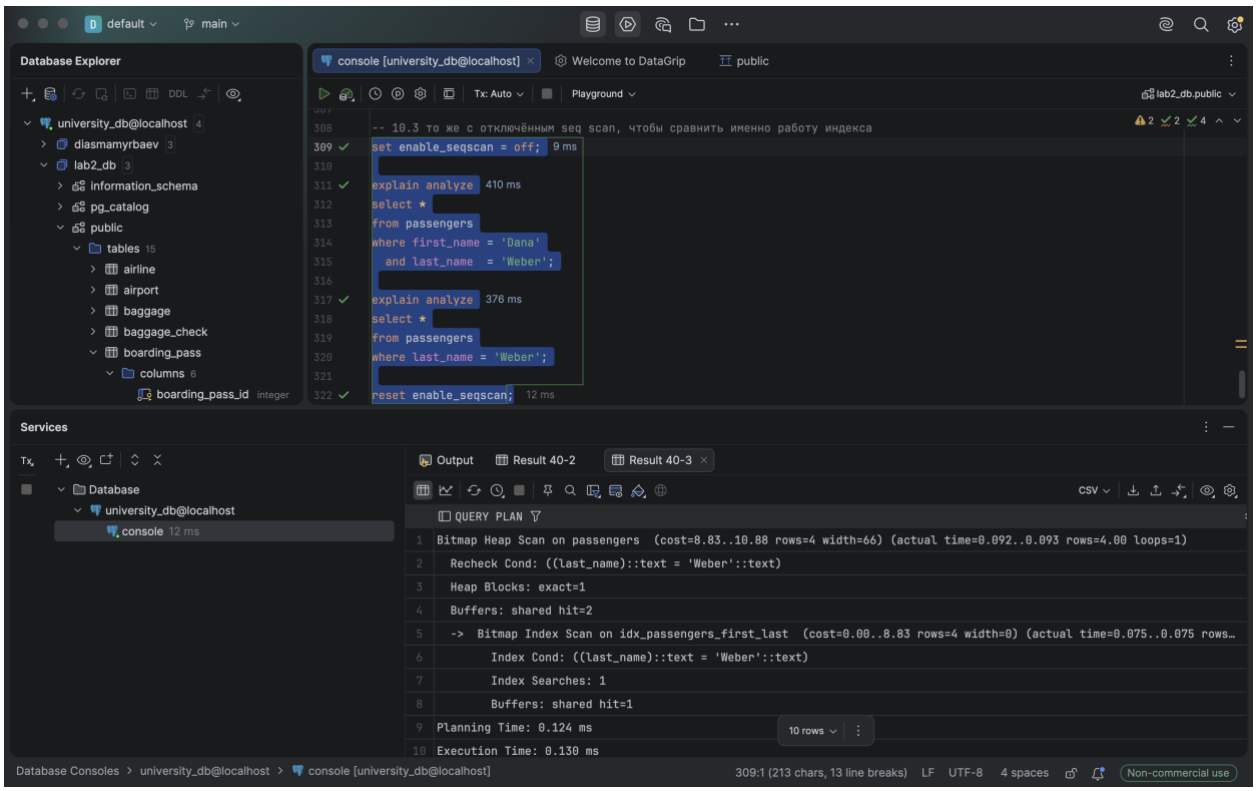
default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables
airline
airport
baggage
baggage_check
boarding_pass
columns
boarding_pass_id integer
booking_id integer
where act_arrival_time is not null
order by act_arrival_time;
-- на большой таблице flights_perf
create index idx_perf_act_arrival_not_null
on flights_perf (act_arrival_time)
where act_arrival_time is not null;
analyze flights_perf;
explain analyze 389 ms
select flight_id, act_arrival_time
from flights_perf
where act_arrival_time is not null
and act_arrival_time >= timestamp '2025-02-01 00:00'
and act_arrival_time < timestamp '2025-02-02 00:00';
Services
Database
university_db@localhost
console 389 ms
Output
Result 35
CSV
QUERY PLAN
Index Scan using idx_perf_act_arrival_not_null on flights_perf (cost=0.42..49.04 rows=1064 width=12) (actua…
Index Cond: ((act_arrival_time >= '2025-02-01 00:00:00'::timestamp without time zone) AND (act_arrival_tim…
Index Searches: 1
Buffers: shared hit=149 read=6
Planning:
Buffers: shared hit=14 read=1
Planning Time: 1.638 ms
Execution Time: 1.259 ms
8 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
270:1 (214 chars, 5 line breaks)
LF
UTF-8
4 spaces
Non-commercial use



10. Create a composite index on the Passengers table using first_name and last_name. Write two SQL queries: one filtering by both first_name and last_name, and another filtering only by last_name. Use EXPLAIN ANALYZE to compare their execution plans.




Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
university_db@localhost
diasmamyrbaev
lab2_db
information_schema
pg_catalog
public
tables 15
airline
airport
baggage
baggage_check
boarding_pass
columns 6
boarding_pass_id integer
booking_id integer
last_name. Write two SQL queries: one filtering by both first_name and
last_name, and another filtering only by last_name. Use EXPLAIN ANALYZE
to compare their execution plans.*/
create index idx_passengers_first_last
on passengers (first_name, last_name);
analyze passengers;
-- 10.1 фильтр по обеим колонкам
explain analyze 403 ms
select *
from passengers
where first_name = 'Dana'
and last_name = 'Weber';
Services
Database
university_db@localhost
console 403 ms
Output
Result 38
QUERY PLAN
Seq Scan on passengers (cost=0.00..3.38 rows=1 width=66) (actual time=0.048..0.058 rows=4.00 loops=1)
Filter: (((first_name)::text = 'Dana'::text) AND ((last_name)::text = 'Weber'::text))
Rows Removed by Filter: 88
Buffers: shared hit=2
Planning:
Buffers: shared hit=36 read=1
Planning Time: 0.765 ms
Execution Time: 0.091 ms
8 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
295:1 (94 chars, 4 line breaks) LF UTF-8 4 spaces
Non-commercial use
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
-- 10.1 фильтр по обеим колонкам
explain analyze
select *
from passengers
where first_name = 'Dana'
and last_name = 'Weber';
-- 10.2 фильтр только по last_name
explain analyze 377 ms
select *
from passengers
where last_name = 'Weber';
Services
Database
university_db@localhost
console 377 ms
Output
Result 39
QUERY PLAN
Seq Scan on passengers (cost=0.00..3.15 rows=4 width=66) (actual time=0.079..0.091 rows=4.00 loops=1)
Filter: ((last_name)::text = 'Weber'::text)
Rows Removed by Filter: 88
Buffers: shared hit=2
Planning Time: 0.194 ms
Execution Time: 0.129 ms
6 rows
Database Consoles > university_db@localhost > console [university_db@localhost]
302:1 (67 chars, 3 line breaks) LF UTF-8 4 spaces
Non-commercial use


default
main
Database Explorer
console [university_db@localhost]
Welcome to DataGrip
public
Tx: Auto
Playground
lab2_db.public
university_db@localhost 4
diasmamyrbaev 3
lab2_db 3
information_schema
pg_catalog
public
tables 15
airline
airport
baggage
baggage_check
boarding_pass
columns 8
boarding_pass_id integer
-- 10.3 то же с отключённым seq scan, чтобы сравнить именно работу индекса
set enable_seqscan = off; 9 ms
explain analyze 410 ms
select *
from passengers
where first_name = 'Dana'
and last_name = 'Weber';
explain analyze 376 ms
select *
from passengers
where last_name = 'Weber';
reset enable_seqscan; 12 ms
Services
Database
university_db@localhost
console 12 ms
Output
Result 40-2
Result 40-3
CSV
QUERY PLAN
Bitmap Heap Scan on passengers (cost=8.83..10.88 rows=4 width=66) (actual time=0.092..0.093 rows=4.00 loops=1)
Recheck Cond: ((last_name)::text = 'Weber'::text)
Heap Blocks: exact=1
Buffers: shared hit=2
-> Bitmap Index Scan on idx_passengers_first_last (cost=0.00..8.83 rows=4 width=0) (actual time=0.075..0.075 rows…
Index Cond: ((last_name)::text = 'Weber'::text)
Index Searches: 1
Buffers: shared hit=1
Planning Time: 0.124 ms
Execution Time: 0.130 ms
10 rows
Database Consoles
university_db@localhost
console [university_db@localhost]
309:1 (213 chars, 13 line breaks)
LF
UTF-8
4 spaces
Non-commercial use